Habib University

EE 354 / CE 361 / MATH 310 – Introduction to Probability and Statistics

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Assignment No. 01

Release Date: [04/09/2026]

Due by: [16/09/2026]

Total points: 120

Points obtained:

Student Name:	Student ID:	Section:

EE 354 / CE 361 / MATH 310 – Introduction to Probability and Statistics
Assignment No. 01

Purpose:
The purpose of this assignment is to help you practice constructing sample spaces and events from a described random experiment, applying the axioms of probability, computing probabilities by counting arguments (permutations and combinations), and reasoning about conditional probability, independence, and Bayes' rule.

Instructions:

1. This assignment must be completed individually.
2. All answers must be written in dark ink. Extra sheets may be used.
3. Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

1. Assignments will be graded by the instructor/TA.
2. You must show a reasonable amount of work. Marks may be deducted if important steps are skipped over or missing entirely.
3. You may be called for a viva to demonstrate your understanding. If you cannot correctly answer the question you have attempted right, you may lose your marks.
4. A grade of zero will be awarded if plagiarism is detected.
5. Untidy work may result in a deduction of points.

Late Submission Penalty:

- 1-day late submission - 20% deduction of the maximum allowable marks
- No submission will be accepted after one day of the original deadline

CLO Assessment:
This assignment assesses the following course learning outcomes:

Course Learning Outcomes	**CLO Assessed**
CLO 1: Transform a description of a random experiment into the formal setting of probability theory – construct sample space, identify and specify events. (Cog-4)	✓
CLO 2: Apply the axioms of probability and compute probabilities by counting. (Cog-3)	✓
CLO 3: Compute probabilities of events and moments from given distributions. (Cog-3)	
CLO 4: Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog-3)	
CLO 5: Analyze a random experiment for presence of dependent situations and modeling and analyzing it using notions of conditional probability. (Cog-4)	

Total Marks: 120

Problem Statement 1 [10 Points] - HUCON

Multiverse's flagship anime-con event HUCON's registration system assigns each person request an integer ticket number from 1 to 10, based on a chit they receive at the entry. Sample space: $S = \{1, 2, \ldots, 10\}$. Define events:

- A: ticket number is even
- B: ticket number is greater than 6
- C: ticket number is divisible by 4

(a) Write A, B, C explicitly as subsets of S. [2]

(b) A ticket is flagged as "priority" if it satisfies exactly one of A, B, C. Find the set of priority tickets. [2]

(c) A ticket is "Free ki" if it satisfies neither B nor C. Express this using complements and find the subset. [2]

(d) Verify whether $(A \cap B) \cup (A \cap C) = A \cap (B \cup C)$ using this specific example, and then explain why this identity holds in general. [4]

Problem Statement 2 [10 Points] - Gacha Gamer

Reeba is a die hard fan of gacha games and after grinding hours on one of them she finally unlocked a gacha style loot box. A gacha-style loot box gives one of 5 items each time you open it, numbered 1 to 5. Item 5 (the "rare" item) is 3 times as likely to drop as any of the other 4 items, and items 1 through 4 are all equally likely to each other.

(a) Find the probability law, i.e., $P(i)$ for each $i = 1, \ldots, 5$. [3]

(b) What is the probability of getting an item numbered 3 or higher? [3]

(c) Reeba claims "items 1 through 4 together are less likely than item 5 alone." Verify or refute this using your probabilities. [4]

Problem Statement 3 [10 Points] - CE Major AT

Hellooouuu, after giving his A game during his 4 years of university, CE Major AT gets hired in a quality control plant. A quality control line inspects items and classifies each into exactly one of the categories: {Perfect, Minor Defect, Major Defect, Scrap}, forming a partition of the sample space. It is known that $P(\text{Perfect}) = 0.7$, $P(\text{Minor Defect}) = 0.2$, and Major Defect is twice as likely as Scrap.

(a) Find $P(\text{Major Defect})$ and $P(\text{Scrap})$. [3]

(b) An item is "acceptable" if it is Perfect or has a Minor Defect. Find $P(\text{Acceptable})$ using the additivity axiom, explicitly stating why it applies. [3]

(c) AT being the visionary he is, comes up with a new independent inspection line which has $P(\text{Acceptable}) = 0.85$. If a batch is drawn from one of the two lines with equal probability, is "which line the batch came from" together with "whether the item is acceptable" something you can determine using the axioms alone, or do you need more tools? Explain briefly (conceptual, no calculation needed). [4]

Problem Statement 4 [10 Points] - Biometric Successful

HU security team uses fingerprint scanners at the Gate 1 and Gate 3 entrances. If a person is authorized, the scanner correctly grants access 98% of the time. If a person is unauthorized, the scanner incorrectly grants access 2% of the time (a false accept). It is known that 99.9% of people attempting entry are authorized.

Given that a person was granted access, find the probability that they are actually unauthorized. Show your full tree diagram or Bayes' rule computation.

Problem Statement 5 [10 Points] - Swift vs City

Ahmed enjoys roaming in his Red Swift in 3 major zones: Zone A (50% of all drives), Zone B (30%), and Zone C (20%). The probability that a long drive plan is cancelled by Ahmed is 0.02 in Zone A, 0.05 in Zone B, and 0.10 in Zone C.

(a) Find the overall probability that a random long drive is cancelled. [2]

(b) Given that a long drive was cancelled, find the probability it was supposed to be in Zone C. [2]

(c) Ahmed's mother claims that "whether a long drive is cancelled" is independent of "which zone it was supposed to be in." Using your results from part (a) and the cancellation rates given, verify or refute this claim. [3]

(d) Sheeru, a notorious local racer in his modified Honda City, is always on the lookout for Ahmed. If Ahmed's drive is **not** cancelled, Sheeru spots him and initiates a race with probability 0.4 in Zone A, 0.7 in Zone B, and 0.9 in Zone C. Due to varying traffic conditions, Ahmed's probability of winning the race against Sheeru is 0.2 in Zone A, 0.5 in Zone B, and 0.8 in Zone C. Given that Ahmed went out for a drive and eventually won a race against Sheeru, find the probability that this victory occurred in Zone B. [3]

Problem Statement 6 [10 Points] - Highkey Brownies

In a game show, Hamza, a contestant, is shown 4 identical boxes. One contains a Delicious Brownie; the other 3 are empty. Hamza picks Box 1. The host, who knows where the prize is, then opens two of the remaining three boxes, always revealing empty ones, and offers Hamza the choice to switch to the one remaining unopened box.

(a) Compute P(brownie is in Hamza's original box) before any boxes are opened. [2]

(b) After the host opens two empty boxes (from the remaining three), compute P(original box | host's action), the probability the brownie is in Hamza's original box given the host's action. Also compute P(remaining box | host's action), the probability it is in the remaining unopened box. [5]

(c) Should Hamza switch? Justify using conditional probability. Intuition alone will not receive credit. [3]

Problem Statement 7 [10 Points] - Munir XI Returns

Munir Tariq aka Mnt, after giving his debut in Munir XI, and displaying a generational performance against Babar Azam XI, looks forward to a new venture. Mnt now wishes to explore the Pakistan Stock Exchange, and as usual after leaving his class in the middle, he decides to monitor stocks of the company SA International. The stock's price moves up by one unit with probability $p = 0.6$ each day, or down by one unit with probability 0.4, independently each day, starting at 0.

(a) Find the probability that after 4 days, the price is exactly at 0. [3]

(b) Find the probability that after 3 days, the price is at +1. [3]

(c) Given that the price is at +1 after 3 days, find the probability that the first day's move was up. [4]

Problem Statement 8 [10 Points] - Independence Trap

Bilel mistook "conditional independence" explained in his PNS class for financial independence. He tosses two independent fair coins. Let H_1 = "first coin is Heads," H_2 = "second coin is Heads," and D = "the two coins show different results."

(a) Are H_1 and H_2 independent? (Should be immediate.) [2]

(b) Are H_1 and H_2 independent given that D occurred? Compute $P(H_1 \cap H_2 \mid D)$, $P(H_1 \mid D)$, and $P(H_2 \mid D)$ explicitly and compare. [5]

(c) Explain in one or two sentences why conditioning on D changes the independence relationship. [3]

Problem Statement 9 [10 Points] - Academic Affair

Starting this semester, the university has rolled out a new state-of-the-art facial-recognition attendance system for all classes. When a student marks attendance, the system scans their face and either accepts or rejects the marking.

Testing of the system shows the following. If the person marking attendance is genuinely the enrolled student (a "Present" marking), the system correctly accepts them 95% of the time (and rejects them, prompting a manual override, the remaining 5% of the time). If the person marking attendance is actually a friend attempting proxy attendance on someone else's behalf (a "Proxy" marking), the system incorrectly accepts them 3% of the time (and correctly rejects them 97% of the time). Across the whole course, it is estimated that only 1% of all attendance markings are proxy attempts.

(a) Draw a full probability tree for this scenario, showing both first-stage branches (Present, Proxy) and both second-stage branches (Accepted, Rejected) under each. Label every branch with its probability. [3]

(b) Find the probability that a randomly selected attendance marking is accepted by the system. [2]

(c) Given that a marking was accepted, find the probability that it was actually a proxy attempt. [3]

(d) The course coordinator claims, "if the system accepts a marking, we can be at least 95% sure it is genuine." Using your answer to part (c), verify or refute this claim numerically. [2]

Problem Statement 10 [10 Points] - RAE Shenanigans

RAE, being the most "hardworking" cabinet among all the cabinets, decided to take on the task of selecting 4 people from a pool of 7 faculty and 5 students to form a review panel.

(a) If the panel has no distinguished roles (everyone serves equally), how many different panels are possible? Justify why this is a combination and not a permutation. [3]

(b) If the panel of 4 must additionally elect a Chair and a Vice Chair from among the 4 selected (2 distinct roles, one person each), how many total arrangements are possible? [3]

(c) How many panels of 4 are possible if the panel must include at least 2 faculty members? [4]

Problem Statement 11 [10 Points] - Faryaad The Band

After delivering a banger performance at the HU concert, the HU band receives a hefty cheque and decides to spend it on some new equipment. Ali Mahmoodi has been given the responsibility to oversee this huge order so that nothing goes wrong. A shipment of 20 electronic guitars contains 5 defective units and 15 functioning units. Ali randomly selects 6 guitars for testing (without replacement, order does not matter).

(a) Find the probability that exactly 2 of the selected guitars are defective. [5]

(b) Find the probability that at least 1 selected guitar is defective. [5]

Problem Statement 12 [10 Points] - Cantt View Apt No: 110

In Cantt View apt 110 there is a long-ongoing tradition of playing different card games in the evening. One day, the Cantt View boys decide to test their PNS knowledge on those cards. Recall that a standard deck of playing cards has 52 cards total. The deck is divided into 4 suits: hearts, diamonds, clubs, and spades. Hearts and diamonds are red; clubs and spades are black. Each suit has 13 cards, one of each rank: Ace, 2, 3, ..., 10, Jack, Queen, King. This means there are exactly 4 cards of each rank (one per suit), for example, there are 4 Aces, one in each suit.

A hand of 5 cards is dealt from a well-shuffled deck, without replacement, and the order in which the cards are dealt does not matter.

(a) How many different 5-card hands are possible in total? [2]

(b) If instead the order in which the 5 cards were dealt *did* matter (i.e., you cared which card came first, second, and so on), how many ordered 5-card deals would be possible? [2]

(c) A hand is called a *full house* if it contains exactly 3 cards of one rank and exactly 2 cards of a different rank (for example, three 7s and two Kings). Find the probability that a randomly dealt 5-card hand is a full house. [3]

(d) A hand is called a *flush* if all 5 cards are from the same suit (the ranks do not need to be consecutive). Find the probability that a randomly dealt 5-card hand is a flush. [3]